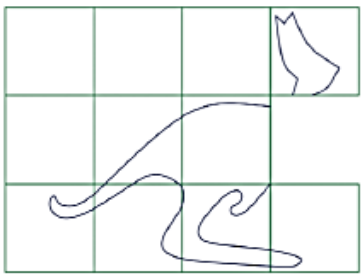
3 Points Each

2002

1. Which of the squares below should be put into the picture to get the symbol of our competition?



a) 

b) 

c) 

d) 

e) 

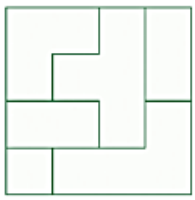
2. After we simplify
2 + 2 – 2 + 2 – 2 + 2 – 2 + 2 – 2 + 2
the result will be:

a) 0
b) 2
c) 4
d) 12
e) 20

3. Andrzej received three cars, four balls, three teddy bears, ten pens, two chocolate bars, and a book for his birthday. How many items did he get in all?

a) 15
b) 17
c) 20
d) 23
e) 27

4. A square was divided into pieces (see the picture). Which of the pieces below does not occur in this divided square?



a) 

b) 

c) 

d) 

e) 

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5. Gina, Lena, Suzie, and Joanna have their birthdays on March 1st, May 17th, July 20th, and March 20th (not necessarily in that order). Lena and Suzie were born in the same month. Gina and Suzie were born on the same day of the month. Which of the girls was born on May 17th?

a) Gina
b) Lena
c) Suzie
d) Joanna
e) This cannot be determined from the information given.

6. The human heart beats an average of 70 times per minute. On average how many times does it beat during one hour?

a) 42, 000
b) 7,000
c) 4,200
d) 700
e) 420

7. Quadrilateral ABCD is a square and its side is 10 cm long. Quadrilateral ATMD is a rectangle and its shorter side is 3 cm. What is the difference between the sum of the lengths of all the sides of the square and the sum of the lengths of all the sides of the rectangle ATMD?

a) 14 cm
b) 10 cm
c) 7 cm
d) 6 cm
e) 4 cm

8. Which of the figures below couldn't be made by folding a rectangular sheet just once (see the picture)?



a)
b)
c)
d)
e)



4 Points Each

9. The houses on the street where John lives are numbered from 1 to 24. How many times does the digit 2 appear in the numbering of these houses?

a) 2
b) 4
c) 8
d) 16
e) 32

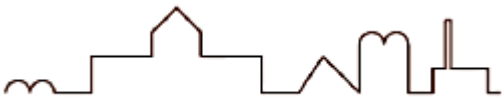
10. There are six identical oranges on one side of the scale and two identical melons on the other side of the scale. After we add one melon to the side with the oranges, the scale will be balanced. How many oranges weigh as much as one melon?



a) 2
b) 3
c) 4
d) 5
e) 6

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11. The picture below is a sketch of a castle. Which of the lines below is not part of the sketch?



a) 

b) 

c) 

d) 

e) 

12. We add 17 to the smallest two-digit number and then we divide the sum by the largest one digit number. What is the result?

a) 3
b) 6
c) 9
d) 11
e) 27

13. In a certain ancient country the numbers one, ten, and sixty were expressed with the following symbols:



People were writing down other numbers using these symbols. For example, the number 22 was written as:



Which of the following notations represents the number 124?

a) 

b) 

c) 

d) 

e) 

14. The face of a clock was divided into four parts. The sums of the numbers in each of those parts are consecutive numbers. Which of the following pictures satisfies this rule?


a)


b)


c)


d)

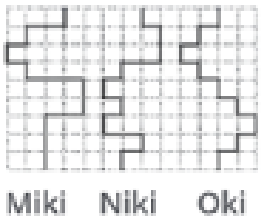

e)

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15. Klara and Zoe had 60 matches altogether. Klara took as many matches as she needed to build a triangle with each side 6 matches long. Zoe used the remaining matches to build a rectangle which had one side equal to 6 matches. How many matches did she use to make one longer side of this rectangle?

a) 9
b) 12
c) 15
d) 18
e) 30

16. Three kangaroos, Miki, Niki, and Oki, participated in a competition. Jumping at the same speed, they jumped along the lines you can see in the picture. Only one of the sentences A, B, C, D, and E listed below is true. Which one is it?



a) Miki and Oki finished at the same time.
b) Niki finished first
c) Oki finished last
d) All the kangaroos finished at the same time
e) Miki and Niki finished at the same time.

5 Points Each

17. Each boy, Mike, Nate, Oliver, and Paul, has exactly one of the following animals: a cat, a dog, a goldfish, and a canary. Nate has a pet with fur. Oliver has a pet with four legs. Paul has a bird, and Mike and Nate don't like cats. Which of the following sentences is not true?

a) Oliver has a dog.
b) Paul has a canary.
c) Mike has a goldfish
d) Oliver has a cat
e) Nate has a dog

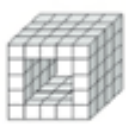
18. Mary leaves her house at 6:55 and arrives at school at 7:32. Zoe needs 12 minutes less than Mary to get to school. Yesterday Zoe showed up at school at 7:45. What time did she leave her house?

a) at 7:07
b) At 7:20
c) At 7:25
d) At 7:30
e) At 7:33

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19. Robert had a certain number of identical cubes. He made a tunnel using half of his blocks (see Picture 1). With some of the remaining cubes he formed a pyramid (see Picture 2). How many blocks were left after making both of these structures?

a) 34
b) 28
c) 22
d) 18
e) 15


Picture 1


Picture 2

20. The daughter is 3 years old, and her mom is 28 years older than the daughter. How many years later will the mom be three times as old as her daughter?

a) 9
b) 12
c) 10
d) 1
e) 11

21. A conductor wanted to make a trio consisting of a violinist, a pianist, and a drummer. He had to choose one of two violinists, one of two pianists, and one of two drummers. He decided to try each of the possible trios. How many attempts did he have to make?

a) 3
b) 4
c) 8
d) 24
e) 25

22. One medal can be cut out from a golden square plate. If four medals are made from four plates, the remaining parts of those four plates can be used to make one more plate. What is the largest number of medals that can be made when 16 plates are used?

a) 17
b) 19
c) 20
d) 21
e) 32

23. Twenty eight students from the fourth grade competed in a math competition. Each student earned a different number of points. The number of students who received more points than Tom is two times smaller than the number of students who had fewer points than Tom. In which position did Tom finish that competition?

a) 6th
b) 7th
c) 8th
d) 9th
e) 10th

24. An odometer in a car shows the number 187569, which is the number of kilometers that have been traveled. This number consists of all different digits. After traveling how many more kilometers will the odometer show a number consisting of all different digits again?

a) after 777 km
b) after 12, 431 km
c) After 431 km
d) After 21 km
e) After 11 km

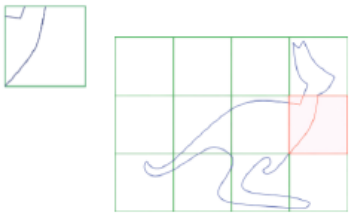
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Answers key:

1. B
2. C
3. D
4. D
5. D
6. C
7. A
8. B
9. C
10. E
11. C
12. A
13. E
14. B
15. C
16. E
17. A
18. B
19. D
20. E
21. C
22. D
23. E
24. D

3 Point Solutions

1 (B)



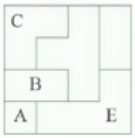
2 (C) 4

$2 + 2 - 2 + 2 - 2 + 2 - 2 + 2 - 2 + 2 =$
$= 2 + 0 + 0 + 0 + 0 + 0 + 2 = 2 + 2 = 4$

3 (D) 23
Just add the numbers of gifts Andrzej received: 3 + 4 + 3 + 10 + 2 + 1 = 23.

4 (D)

The picture shows the figures that can be found in the square.

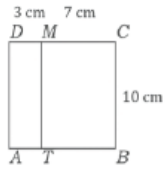


5 (D) Joanna
Lena and Suzie were born in the same month, so they were born in March. Gina and Suzie were born on the same day of the month, so their birthdays fall on the 20th. This leaves Joanna as the girl who was born on May 17th.

6 (C) 4,200
There are 60 minutes in an hour, so during one hour the human heart beats on average 60 × 70 = 4200 times.

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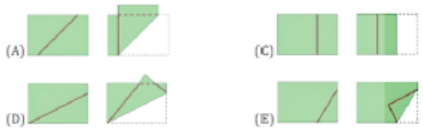
7 (A) 14 cm
The lengths of two of the sides of the square *ABCD* and the longer sides of rectangle *ATMD* are equal, and the difference in lengths of the other two sides of those figures is 10 cm – 3 cm = 7 cm. The difference between the sum of the lengths of all the sides of the square and the sum of the lengths of all the sides of the rectangle *ATMD* is 2 × 7 cm = 14 cm.



8 (B)



The pictures below show fold lines and edges.



When folding any piece of paper, the result is a shape completely on one side of the folding line, so the red lines (shown to the left) cannot be the folding lines of (B). The purple lines potentially could be the folding lines of (B) but unfolding along any of the purple lines does not produce a rectangle.



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4 Point Solutions

9 **(C) 8**
The digit 2 shows up 8 times, in numbers: 2, 12, 20, 21, 22 (2 used twice), 23, and 24.

10 **(E) 6**
The scale is balanced when there are 2 melons on the one side and 6 oranges plus 1 melon on the other. By removing one melon from each side, we can see that the weight of 6 oranges is equal to the weight of 1 melon.

11 **(C)**



The picture shows where (A), (B), (D) and (E) can be found.



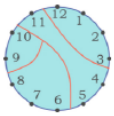
12 **(A) 3**
The smallest two-digit number is 10 and the greatest one-digit number is 9. The sum of 10 and 17 is 27. When 27 is divided by 9 the result is 3.

13 **(E)**



The symbols represent
60 + 60 + 1 + 1 + 1 + 1 = 120 + 4 = 124.

14 **(B)**



1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 + 12 = 78, so the sum of all the numbers on the face of the clock is 78. The only four consecutive numbers for which the sum is 78 are 18, 19, 20, and 21. In picture (B), the sums of the numbers in each part of the clock are: 12 + 1 + 2 + 3 = 18; 10 + 9 = 19; 11 + 4 + 5 = 20; 6 + 7 + 8 = 21. For each of the other clocks we can find a part with the sum of its numbers less than 18.

15 **(C) 15**
Klara built a triangle using 6 × 3 = 18 matches. There were 60 – 18 = 42 matches left for Zoe.

The length of one of the sides of the rectangle is equal to 6 matches. So, Zoe used 6 + 6 = 12 matches to build that side and the one opposite. The number of matches left for the other two sides is 42 – 12 = 30. So, each of those sides is made out of half of the 30 matches, which is 15.

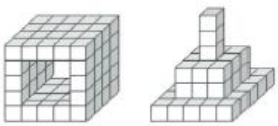
16 **(E) Miki and Niki finished at the same time.**
The lengths of the routes taken by the kangaroos are: Miki, 18 units; Niki, 18 units; and Oki, 17 units. Because they were jumping with equal speeds, Miki and Niki finished at the same time because their routes were the same length.

5 Point Solutions

17 **(A) Oliver has a dog.**
Nate, who has a pet with fur but doesn't like cats, has to have a dog. Because Nate owns a dog, the sentence "Oliver has a dog" is not true. Actually, Oliver owns a cat, and Nate has the dog. The other sentences are true.

18 **(B) at 7:20**
Mary needs 37 minutes to get to school so Zoe needs 37 – 12 = 25 minutes. Since Zoe arrived at school at 7:45, she had to leave her house 25 minutes earlier, which means she left at 7:20.

19 **(D) 18**



Robert made a tunnel using

5 × 5 × 4 – 3 × 3 × 4 = 100 – 36 = 64 cubes.

This also means that there were 64 cubes left at this point. He made the pyramid out of

5 × 5 + 2 (3 × 3) + 3 = 25 + 18 + 3 = 46 cubes.

Now there were 64 – 46 = 18 cubes left.

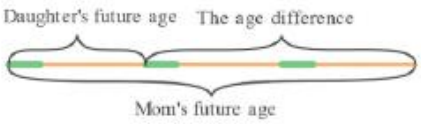
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20 **(E) 11**

The green segment below represents the daughter's age. The orange segment represents the period of time after which her mom's future age is 3 times the daughter's future age.

The daughter's future age is represented by the interval with both colors and the mom's future age is represented by 3 such intervals. The difference between the future ages is still 28 since the age difference is always the same. This age difference is twice the daughter's future age. Thus, half of the age difference is 28 ÷ 2 = 14, so the daughter's future age is 14. She is 3 years old now, so the time period from now to this future point in time is 14 – 3 = 11 years.



21 **(C) 8**

The trio consists of a violinist, a pianist, and a drummer. A violinist alone can be selected in 2 ways. A violinist and a pianist can be selected in 2 × 2 ways since each violinist can be joined by one of 2 pianists. A whole trio is created when a violinist and a pianist are joined by one of 2 drummers, so the conductor can create 2 × 2 × 2 = 8 trios.

22 **(D) 21**

If 4 medals can be made from 4 plates, then from 16 plates we can cut out 16 medals. The remaining material is enough for 4 more plates. From those 4 plates, 4 medals can be cut out and still the remaining material is enough for 1 more plate. From that 1 plate we can have 1 more medal, therefore there are as many as 16 + 4 + 1 = 21 medals which can be made from 16 plates.

23 **(E) 10th**

In addition to Tom, there were 27 other students in the math competition. Since there were twice as many students with fewer points than those with more points than Tom, we can divide 27 into 3 groups, which gives 9 students in each group. Two of the groups are the students who got fewer points than Tom, and one group is the students who got more points. So, there were 9 students who got more points than Tom. Therefore, Tom finished that competition in 10th place.

24 **(D) after 21 km**

We cannot increase just the ones digit since it is already 9. We have to change the last two digits without using 7 or 8. The smallest such two-digit number is 90, so the odometer reading has to be 187590. To reach it the car has to travel 187590 − 187569 = 21 km.

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